加選規則

- 加選優先序:
  - 資工系博士
  - 資工系碩二
  - 資工系碩一
  - 資工系大四
  - 外系研究生
  - 外系大四生
- 有希望加選的同學請於 2/26 (四) 12:00 前寄信至:rogerwu212@gmail.com
- 助教會再回覆同學是否可以加選

Natural Language Processing

Teacher, I have questions

Lecturer:

- Office: science building (理學大樓) 812 room
- Email: albert.yang@nchu.edu.tw

TA:

- 實驗室: science building (理學大樓) 902 room
- Email: rogerwu212@gmail.com

Making an appointment through email in advance is highly recommended!!

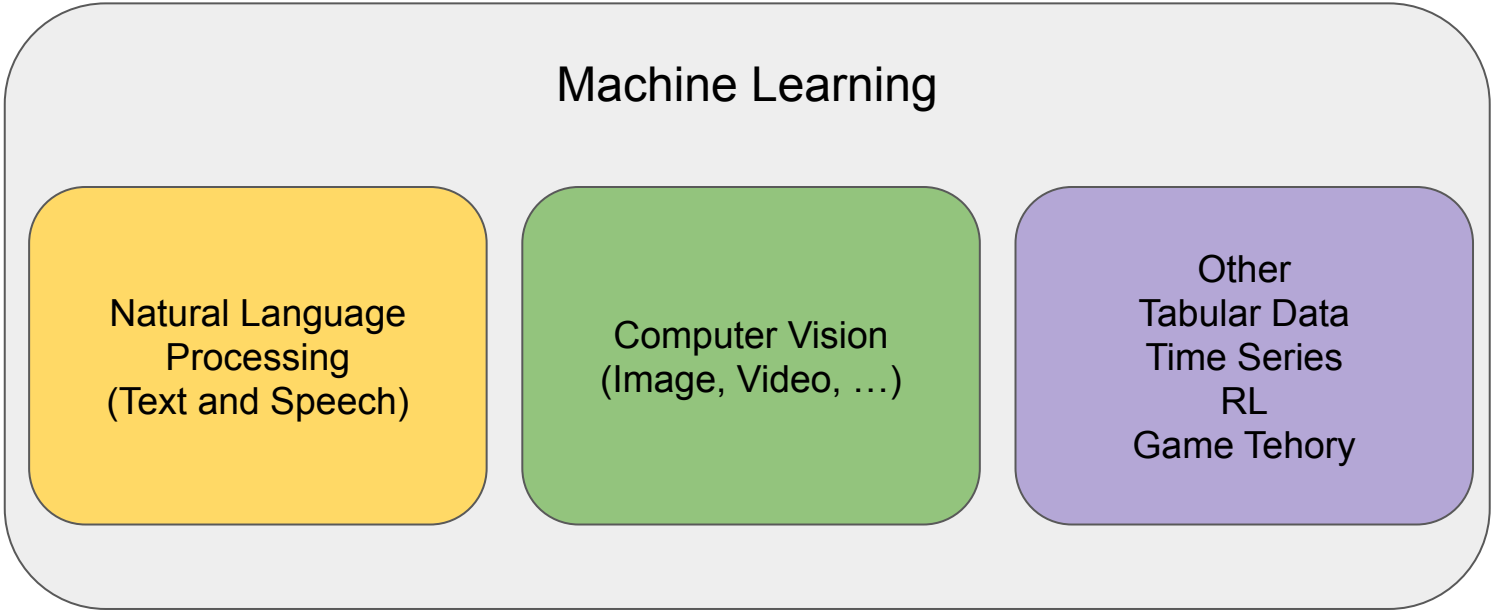
Grading policy

- 期中考 (20%)
- 期末考 (20%)
- 課堂練習 (20%)
- 期末專題（30%）
- 出席 (10%)

Couple of Notes

- Late assignment submissions
  - Submission time will be the basis for grading
  - The maximum score for late submissions is 80 points
  - Do not withdraw the submission at will
- Makeup exam
  - Leave must be completed at least three hours before the class, based on the records in the leave system
  - Must provide valid documents
  - Midterm exam: grades will be combined with the final exam.

What is Natural Language Processing (NLP)



What is Natural Language Processing (NLP)

Prompt: What is NLP?

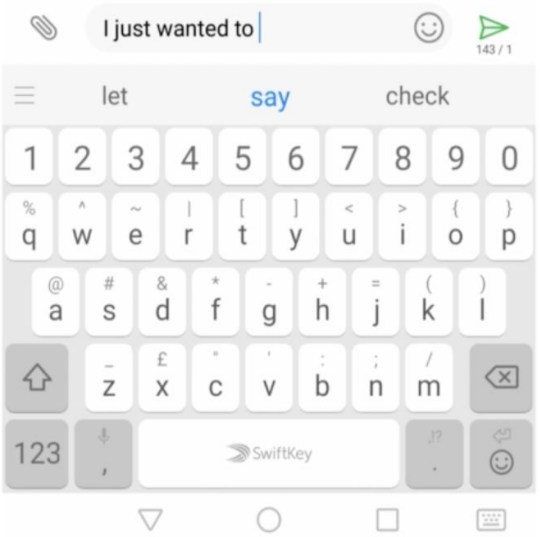
ChatGPT: NLP stands for **Natural Language Processing**. It is a branch of artificial intelligence (AI) that focuses on enabling computers to understand, interpret, and generate human language. NLP combines **linguistics**, **machine learning**, and **computer science** to process and analyze large amounts of natural language data.

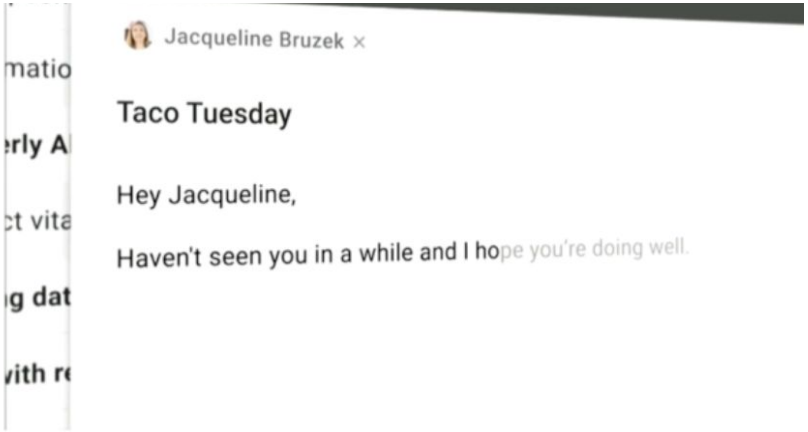
How is NLP Used?

- Spam Detection

How is NLP Used?

- Spam Detection
- Predictive Text



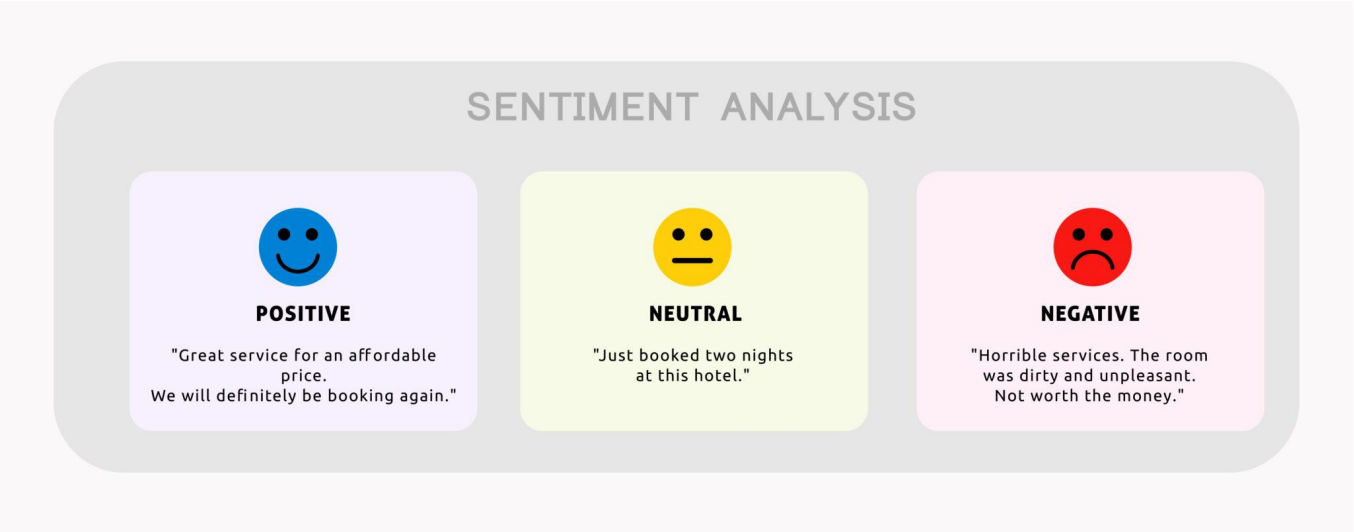


How is NLP Used?

- Spam Detection
- Predictive Text
- Speech Recognition

How is NLP Used?

- Spam Detection
- Predictive Text
- Speech Recognition
- Sentimet Analysis

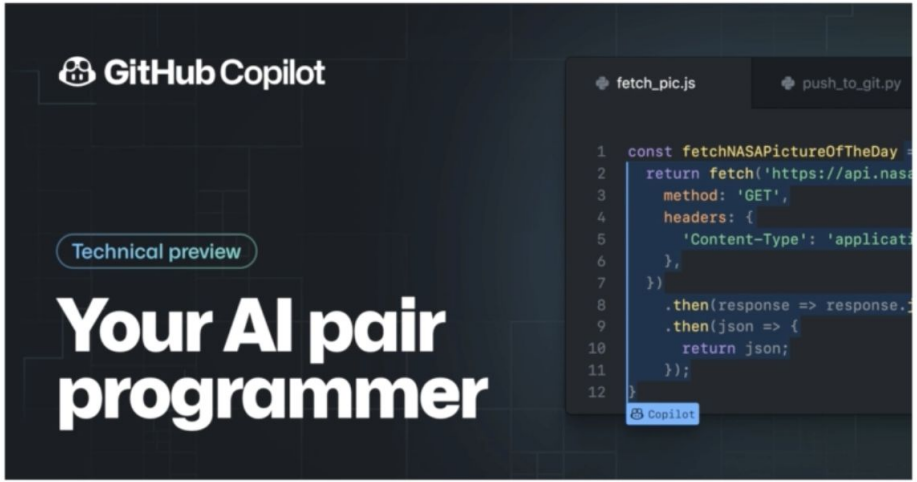


Modern Advancement

- Language Translation

Modern Advancement

- Language Translation
- Text Generation



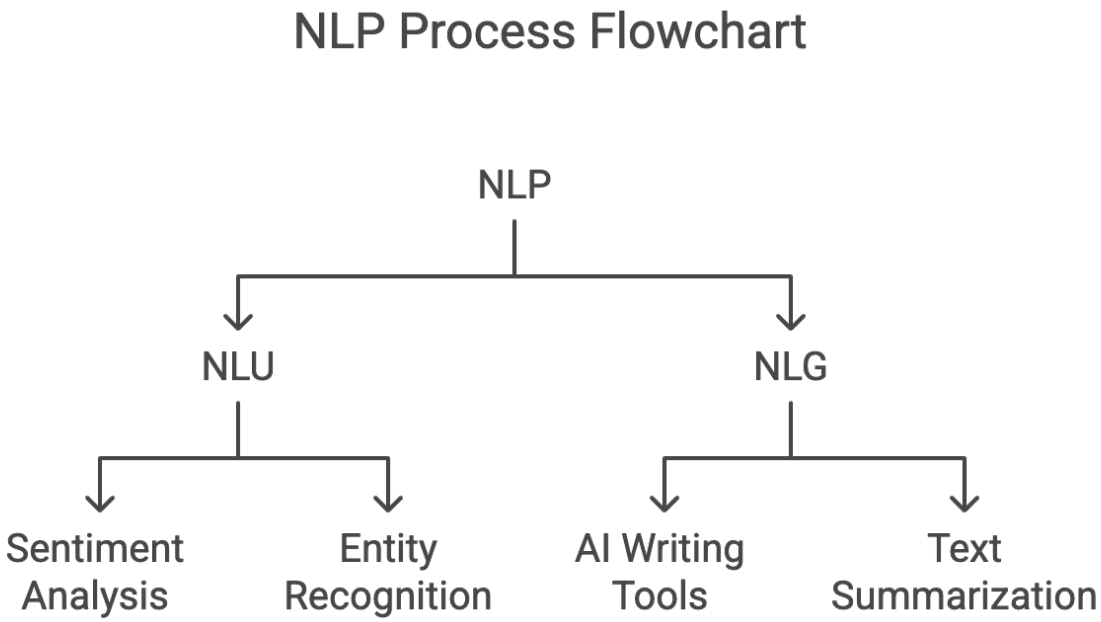
A

What is ChatGPT?

ChatGPT is an AI-powered chatbot developed by OpenAI. It is based on the GPT (Generative Pre-trained Transformer) architecture, specifically GPT-3.5. ChatGPT is designed to engage in text-based conversations with users, providing responses that are coherent, contextually relevant, and generated in a human-like manner.

The underlying technology behind ChatGPT is deep learning, specifically employing a transformer model. This architecture allows the model to process and understand the structure, context, and meaning of text input, enabling it to generate appropriate and informative responses.

How NLP Works?

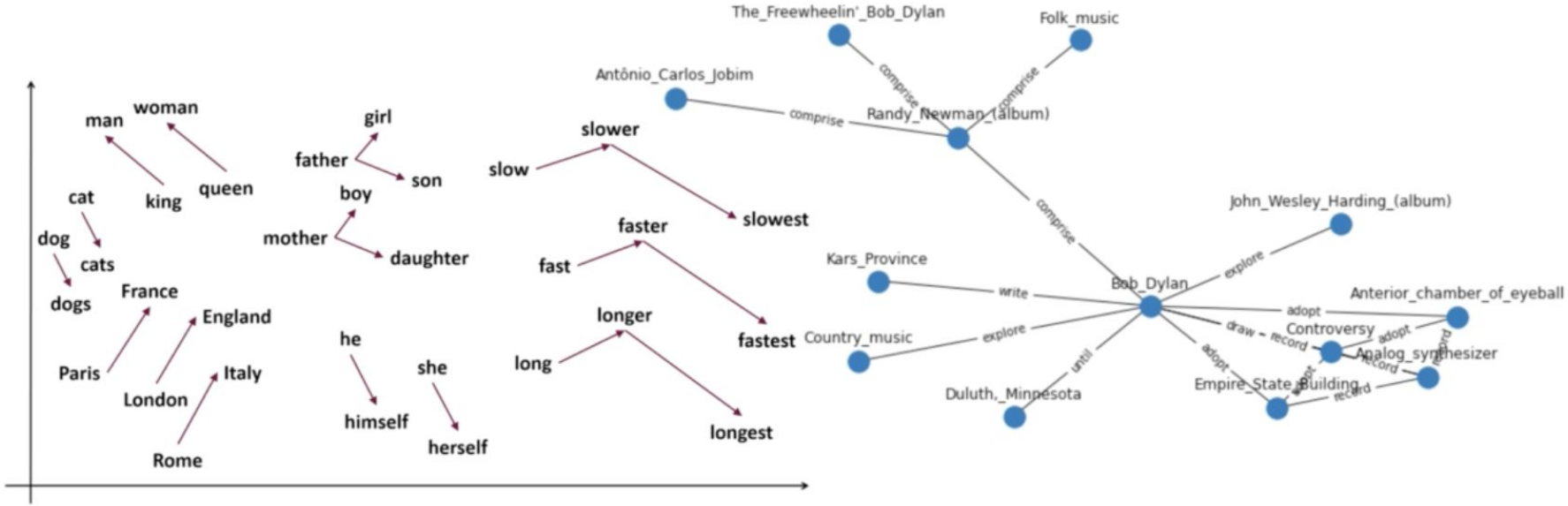


NLP Methods

- Vector-Based Models and Text Processing
- Probability Models
- Machine Learning Models
- Deep Learning Models
- Large Language Models

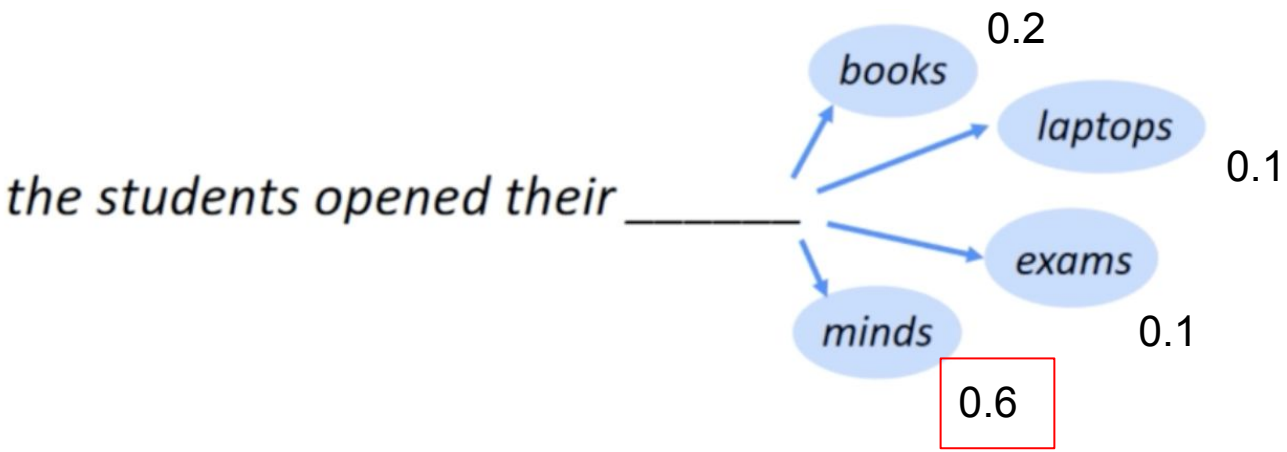
Vector-Based Models and Text Processing

- TF-IDF, Word Embedding Models
- Applications: similarity search, document retrieval, summarization



Probability Models

- Markov Models, N-gram Models, Hidden Markov Models (HMM), Naïve Bayes
- Applications: Spell correction, auto-completion, speech recognition
- Prerequisites for Transformers (BERT and GPT-3)



Machine Learning Models

- Training is essential for ML models

Spam Detection Uses: email filtering, SMS spam detection, phishing detection Method: Naive Bayes, SVM, Random Forest	**Sentiment Analysis** Uses: reputation management, stock market, customer feedback Method: Logistic Regression, SVM, BERT, Random Forest
Latent Semantic Indexing Uses: information retrieval, document clustering, search engines Method: PCA / SVD	**Topic Modeling** Uses: legal document classification, news categorization, recommendation systems Method: Latent Dirichlet Allocation

Deep Learning Models

- RNN, LSTM / GRU, Transformer (BERT, GPT-3)
- Can do any tasks we previously discussed

Can ChatGPT Solve Every NLP Task?

- Specialized tasks require domain-specific models
  - ChatGPT lacks expertise in medical, legal, and financial fields.
  - BioBERT, FinBERT outperform general AI in these areas.

- Privacy & security risks
  - Public cloud-based AI models pose data privacy concerns.
  - Local NLP models (e.g., Llama, Mistral) ensure data control.

- Resource & cost constraints
  - Running ChatGPT is expensive & resource-intensive.
  - Task-specific NLP models are faster, cheaper, and more efficient.

Syllabus (May Change in the Future)

Classic NLP Methods:

- Vector Models and Text Processing
- Probabilistic Models
- Machine Learning Models
- Deep Learning Models
- Large Language Models

NLP Tasks

- Sentiment analysis
- Summarization
- Text Q&A
- Text-to-SQL
- …

Advanced Topics:

- RAG
- AI Agent

Schedule

2026/4/1: no class

2026/4/8: holiday

2026/4/22: midterm exam

2026/6/3: final exam